VIM

8th Place Solution, CURE-BENCH Track 1 (Internal Model Reasoning)

Vladimir Manuilov[1] and Ilya Makarov[1,2,3,4]

Internal Clinical Reasoning

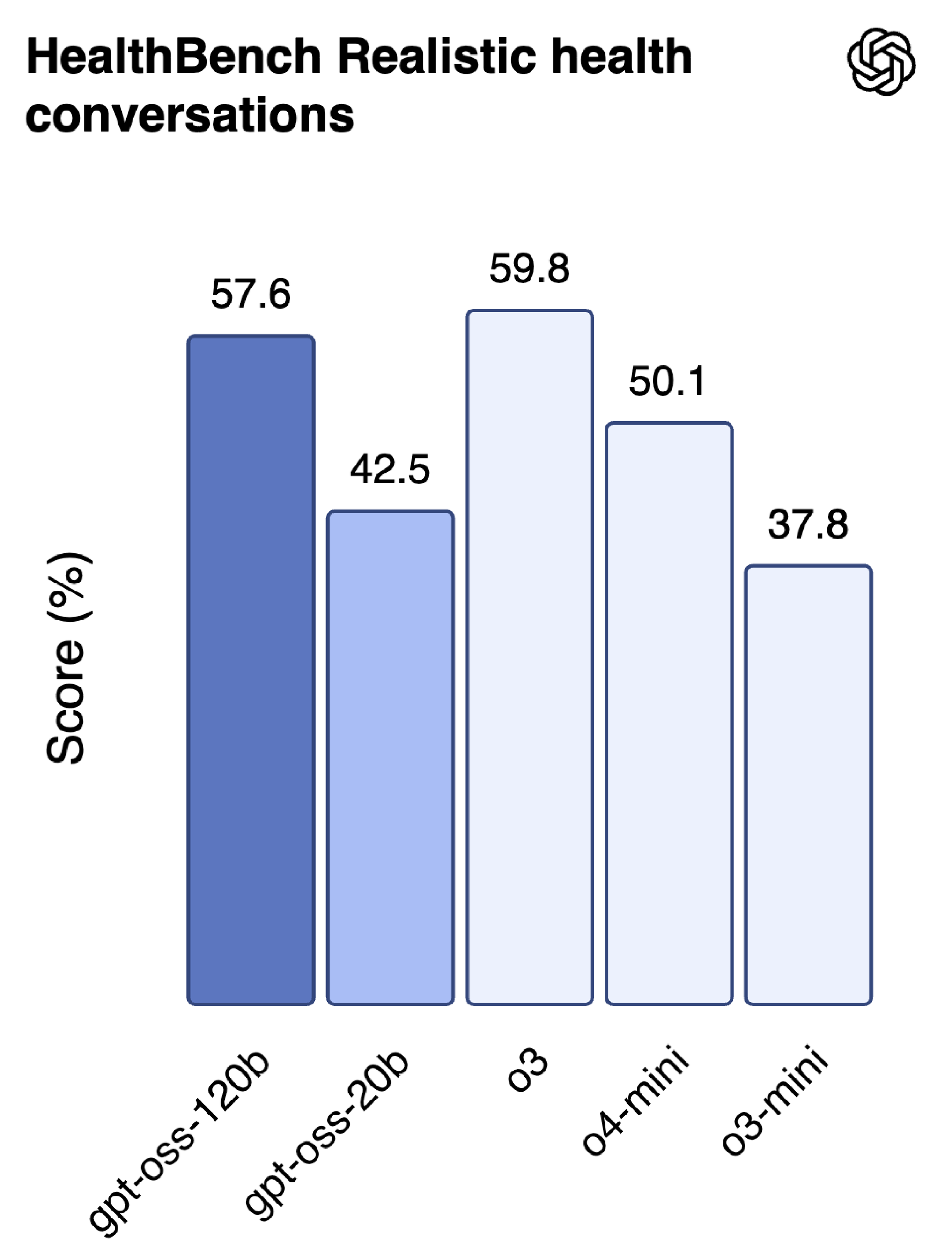
Goal: Evaluate a model's ability to solve complex therapeutic tasks without access to external tools, API, or retrieval.

Scope: 12 biomedical domains, including:

- *Adverse Event Prediction*
- *Drug Warnings and Safety*
- *Dosage and Administration*

Models must generate a detailed **Reasoning Trace**.

Initial Strategy



Hypothesis: Performance on *HealthBench* is a strong predictor of CURE-BENCH reasoning capability.

Baseline: We selected gpt-oss-20b as our starting point based on its high efficiency-to-performance ratio. It achieved **0.34036** on the Public Leaderboard.

Based on HealthBench paper [1], o3 mode was state-of-the-art, followed by Grok 3 and Gemini Pro 2.5. gpt-oss-120b was only worse than o3 based on the model card.

Model Selection

We evaluated a broad spectrum of open-source architectures to find the best reasoning engine:
GLM-4.5, Qwen3-30B-Thinking, DeepSeek-R1, gpt-oss-120b, gpt-oss-20b, etc.

gpt-oss-120b demonstrated superior performance on the open validation set.

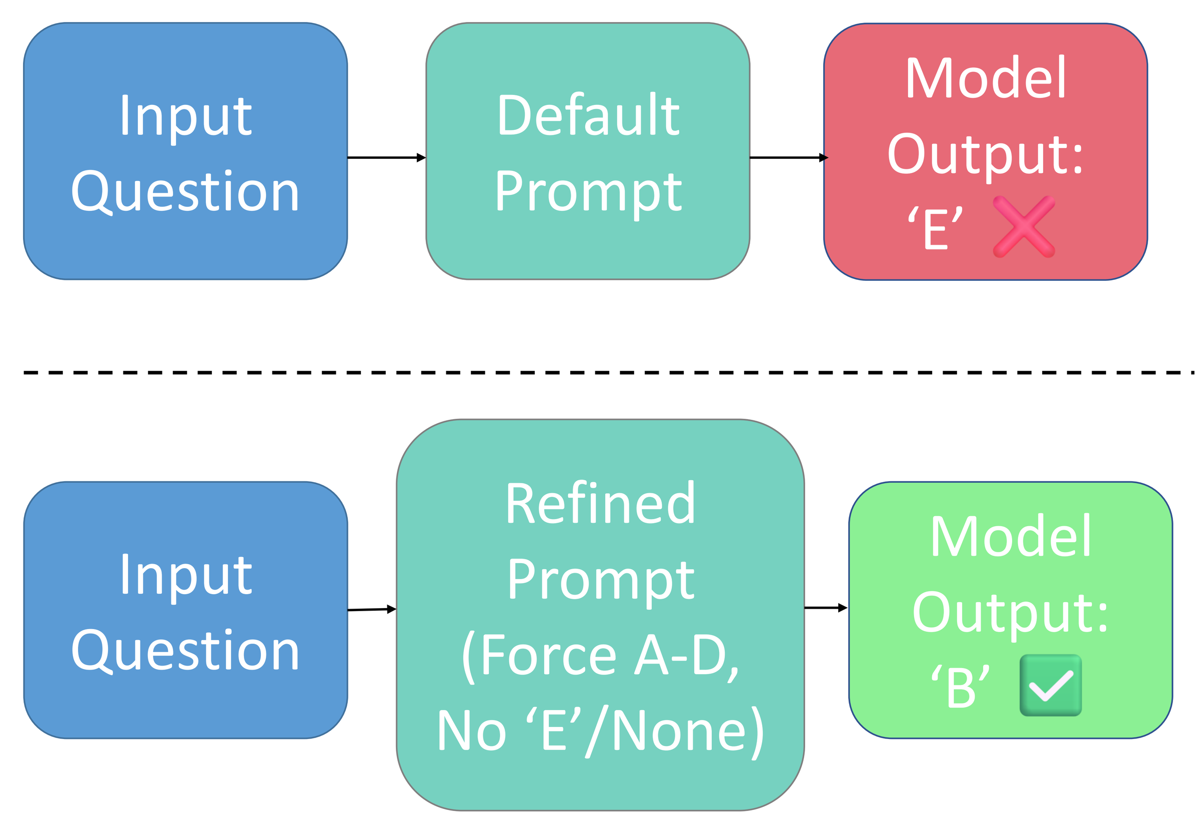
Error Analysis

Observation: Models frequently outputted **"E"** or **"None"** (invalid formats), despite the valid options being strictly A, B, C, or D.

Root Cause: The default prompts explicitly listed 'None' or 'E' as an example option.

Impact: When uncertain, models is prone to chooses options that are 100% incorrect.

Prompt Engineering



- *Removed:* References to 'E' and 'None'.
- *Added:* **"Always choose the closest option even if none is perfect."**
- *Added "System Identity: You are a clinical reasoning assistant... Use standard clinical terminology. Do not invent references."*
- **Increase Reasoning Budget:** gpt-5 has frequently failed to finish reasoning, so we increased it from 1024 to 8192

Results

Using the system prompt and 'constrained choice' for multi-choice and open-ended multi-choice questions, we evaluated closed –sourced gpt-5-mini and gpt-5.

Model	Strategy	Public Score	Private Score
gpt-oss-20b	Baseline (default prompt)	0.34036	-
gpt-oss-120b	Model Selection	0.42598	-
gpt-5-mini	+ Contrained Prompting	0.48687	-
gpt-5 (best)	+ Reasoning Budget	**0.6340**	**0.6801**

Conlusion

What Worked ✅

Forced Choice: Removing "None/E" options forced valid answers via elimination.
Persona Injection: "Clinical Assistant" role reduced refusals and style errors.
High Token Budget: 8k limit enabled deeper Chain-of-Thought reasoning.

What Didn't Work ❌

Small Models: <70B parameter models lacked sufficient medical knowledge.
Complex Prompts: very long prompts did not further boost quality.

1 - Arora, R. K., Wei, J., Hicks, R. S., Bowman, P., Quiñonero-Candela, J., Tsimpourlas, F., ... & Singhal, K. (2025). Healthbench: Evaluating large language models towards improved human health. *arXiv preprint arXiv:2505.08775*.

AIRI

[1]iMak AI Lab,
[2]AIRI, [3]Research Center for AI, Innopolis, [4]ISP RAS
Corresponding author: iam.dadii.dh@gmail.com